Generalized Mach Zehnder Interferometers Integrated on Si_3N_4 Waveguide Platform

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***Abstract*—We demonstrate 3 × 3 and 4 × 4 Generalized Mach Zehnder Interferometers (GMZI) integrated on the Si_3N_4 waveguide platform and present their experimental performance as switching and variable power splitting modules. The GMZI structures have been theoretically modelled by means of a transfer matrix model, evaluating also the GMZI spectral response in the case of fabrication errors that may lead to variations in GMZI arm lengths, MMI phase shifts and MMI splitting ratios. Both 3 × 3 and 4 × 4 GMZI structures were fabricated as integrated Si_3N_4-based circuits and were experimentally characterized with the experimental results being in close agreement with the theoretical predictions. They both reveal a resonant spectral behavior with an FSR of 1.9 nm and low average insertion losses per port that were measured to be 1 ± 0.2 and 2.4 ± 0.4 dB. The two GMZI structures were utilized as 1:3 and 1:4 switching elements, respectively, with average crosstalk levels being always greater than 10 dB for all possible input-output port combinations. Finally, their credentials to serve within linear optical circuit implementations were evaluated through their experimental characterization in variable power splitting applications, showing that the power level at a single output port can take any value between 0 and 100% of the input signal.**

***Index Terms*—Generalized mach-zehnder interferometer, linear optics, optical switches, silicon nitride, silicon photonics.**

I. Introduction

LINEAR optics has turned into a scientific field with a profound range of applications in several modern topics. The employment of linear optical circuits for unitary operations [1], [2], [3], [4] comprises a fundamental technology for quantum photonics [5], with their expansion into universal linear circuitry [6], [7] forming the basis for neuromorphic photonics [8], [9], [10] and microwave photonics [11]. At the same time, switching circuitry extends along the use of linear optics for implementing permutation matrices [12], like for example in the case of the well-known Crossbar and Benes switch architectures [13], [14]. All these application segments have experienced significant advances with the rapid progress witnessed in photonic integration, leading to impressive experimental demonstrations. Photonic integration allowed for the fabrication of low-loss and low-footprint 2 × 2 Mach-Zehnder Interferometers (MZIs) [15], allowing in this way for the deployment of multiple interconnected MZIs in array and mesh layouts towards scaling linear optical systems to practical and highly useful dimensions.

The 2 × 2 MZI structure has been indeed the fundamental building block in all this development roadmap, since it supports a U(2) unitary transformation within a rather simple and mature optical circuit design. However, the MZI-based meshes lead inevitably to differential path losses that stem either from the non-zero MZI losses in the unitary setups or from the need for waveguide crossings in certain switching arrangement, like, for example, in the Benes architecture. This effect gets more pronounced with increasing circuit dimensions, resulting to unitary layouts with reduced fidelity performance and to switch configurations with varying output port losses.

An alternative pathway for overcoming these issues in both unitary and switch layouts could rely on the simplistic approach of generalizing the fundamental building block and replace the 2 × 2 MZI by its generalized counterpart, the $N \times N$ Generalized Mach Zehnder Interferometer (GMZI). Although GMZIs and their credentials to support loss-balanced linear optical circuits were proposed theoretically more than two decades ago [16], their experimental deployment has originally been restricted to SiON-based integration platforms with a single 4 × 4 GMZI requiring a chip area of 12 mm^2 [17]. Establishing its development in a reliable and low loss integration platform, the GMZI could be identified as a highly promising fundamental building block, allowing the realization of high-performance photonic architectures.

In this article we present, for the first time to the best of our knowledge, the fabrication of Si_3N_4-based 4 × 4 and 3 × 3 GMZI and their experimental performance as switching and variable power splitting elements. The 3 × 3 and 4 ×

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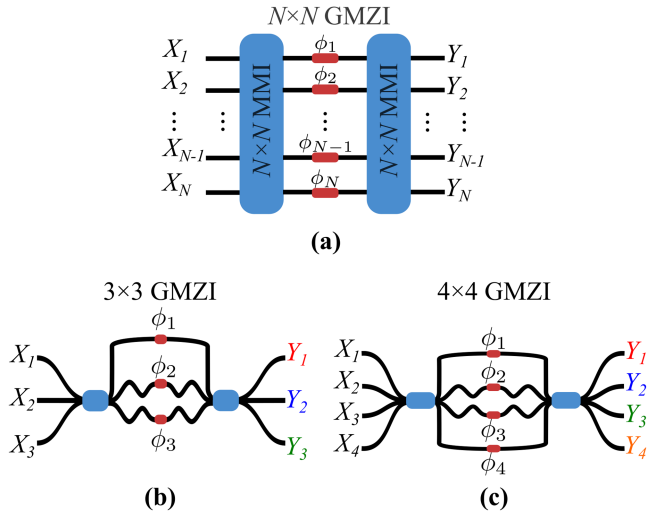


Fig. 1. (a) Block diagram of the $N \times N$ GMZI, consisting of two cascaded MMIs, with each formed arm equipped with a dedicated phase shifter. (b), (c) Designed asymmetric 3×3 and 4×4 GMZI.

4 GMZI designs were theoretically analyzed by means of a generic transfer matrix model, concluding to their theoretically expected spectral behavior when fabrication imperfections lead to arm length differences, multi-mode interferometers (MMI) phase errors and MMI spitting ratio variations. These circuits were then fabricated as integrated Si_3N_4-based devices with footprint requirements not higher than 1.77 mm^2 and 2.7 mm^2, revealing average insertion losses of 1 ± 0.2 and 2.4 ± 0.4 dB per port for the 3×3 and 4×4 layouts, respectively. Their static experimental evaluation was found to be in close agreement with the corresponding theoretically expected performance. Moreover, the two GMZI structures were evaluated as 1:3 and 1:4 switching elements, respectively, with average crosstalk levels being always greater than 10 dB for all possible input-output port combinations. Finally, their credentials to serve within linear optical circuit implementations were validated through their experimental characterization in variable power splitting applications, showing that the power level at a single output port can take any value between 0 and 100% of the input signal.

II. Theory of Generalized Mach Zehnder Interferometers

The $N \times N$ GMZI is illustrated in Fig. 1(a), and consists of two cascaded $N \times N$ MMIs, connected by an array N of waveguides, with every i-th waveguide having a length of L_i. An independent phase shifter ϕ_i is attached to each arm of the GMZI, allowing the tunability of the device. The input optical fields of the GMZI can be defined as a column vector $\boldsymbol{X} = [X_1, X_2, ..., X_N]$, which for the case of single input operation contains a single non-zero element.

The output optical fields of the GMZI $\boldsymbol{Y} = [Y_1, Y_2, ..., Y_N]$ are calculated by multiplying the input fields $\boldsymbol{X}$ with the transfer matrix of the GMZI $\boldsymbol{T}$.

$$\boldsymbol{Y} = \boldsymbol{T} \times \boldsymbol{X}\,. \tag{1}$$

The transfer matrix of the GMZI is an $N \times N$ unitary matrix, defined as:

$$\boldsymbol{T} = \boldsymbol{S} \times \boldsymbol{\Phi} \times \boldsymbol{S}. \tag{2}$$

where $\boldsymbol{S}$, $\boldsymbol{\Phi}$ are the $N \times N$ transfer matrices of the MMIs and the interleaved waveguide array respectively. In particular, $\boldsymbol{S}$ is defined as:

$$\boldsymbol{S} = \frac{\boldsymbol{V}}{\sqrt{N}} \tag{3}$$

with $\boldsymbol{V}$ being a symmetric complex matrix with matrix elements $V_{jk} = \exp(i\theta_{jk})$. The respective arguments θ_{jk} can be theoretically calculated [17], [18] and simplified to a compact expression:

$$\theta_{jk} = -\frac{\pi}{2}(-1)^{j+k+N} + \frac{\pi}{4N} \left[j + k - j^2 - k^2 + (-1)^{j+k+N} \left(2jk - j - k + \frac{1}{2} \right) \right] \tag{4}$$

For ideal and lossless waveguides, $\boldsymbol{\Phi}$ is a diagonal matrix with matrix elements:

$$\Phi_{kk} = \exp\left\{ i \left[\left(\frac{2\pi n_{eff}}{\lambda} \right) L_k + \phi_k \right] \right\} \tag{5}$$

where n_{eff} is the effective refractive index of the waveguide, λ is the wavelength of the optical signals, L_k is the length of the k-arm of the GMZI and ϕ_k is the induced phase shift of the k-phase shifter. Performing the above matrix operations allows the calculation of the optical fields at the output ports of the GMZI for any arbitrary design or device configuration, and consequently the respective normalized output power $T_k = |Y_k|^2 / |\boldsymbol{X}|^2$.

The GMZI devices that were designed on the Si_3N_4 platform followed a 3×3 and 4×4 configuration, as shown in Fig. 1(b) and (c). The lengths of the interleaved waveguides are set to $L_1 = 2337.759$ um, $L_2 = 1709.538$ um and $L_3 = 1709.338$ um for the 3×3 GMZI, and $L_{1,4} = 1924.159$ and $L_{2,3} = 1295.838$ um for the 4×4 GMZI. In the wavelength range of interest the effective refractive index has been set to 1.98 for Si_3N_4 waveguides.

The theoretically expected performance of the Si_3N_4 fabricated structures when launching light through a single input port can be obtained by using (1)–(5). For lossless 4×4 and 3×3 GMZIs with inactive phase-shifters the theoretical output spectra are presented in Fig. 2. Fig. 2(a) and (b) depict the output spectra for the 4×4 GMZI when light is launched only through input port $j = 1$ or $j = 2$, while Fig. 2(c) and (d) illustrate the respective outputs for the 3×3 GMZI. As can be observed, both the 4×4 and 3×3 configurations yield a resonant output with its free spectra range (FSR) value being defined by the differential length employed between the GMZI phase arms. In particular, it is expected that for the introduced arm length differences $\delta x'$ in the range 600 um, common for both devices, the expected FSR will be approximately equal to $\lambda^2/(n_g \delta x') \approx 1.9$ nm. An optimized circuit design could allow even for a zero-length difference that would correspond to a non-resonant broadband behavior, but in our case this length difference was mainly dictated by the need to fit our design within the limited MPW space

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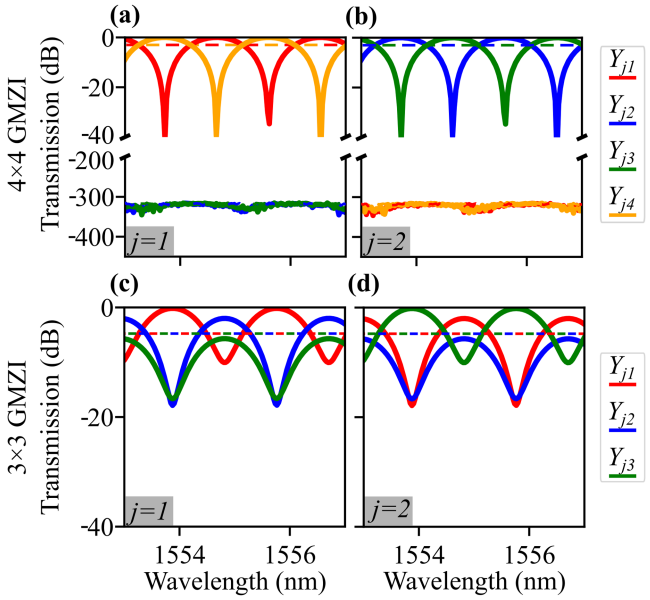


Fig. 2. Output spectra of ideal GMZIs, for the case of balanced (dashed lines) and imbalanced devices (solid lines). The results corresponding to the indicative cases of single inputs X_1, X_2 are presented in (a), (b) for the 4 × 4 GMZI and (c), (d) for the 3 × 3 GMZI respectively.

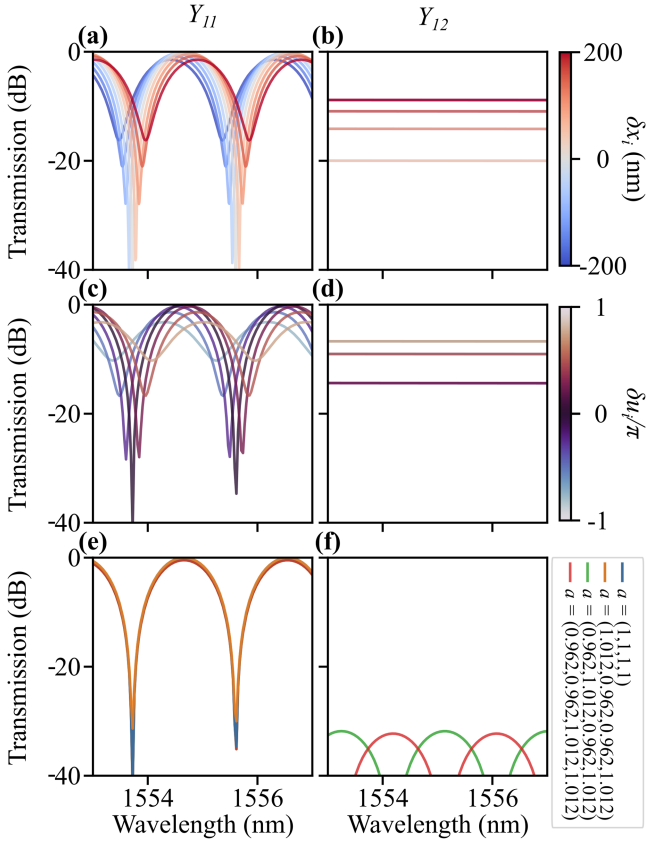


Fig. 3. Effect of characteristic variations to an imbalanced 4 × 4 GMZI. (a), (b): The effect of introducing arm length variations δx_i to the arm 1 of the device for the output spectra Y_{11}, Y_{12} respectively. (c), (d): The effect of introducing MMI phase variations δu to the arm 1 of the device for the output spectra Y_{11}, Y_{12} respectively. (e), (f): The effect of splitting ratio variations δa at the output spectra Y_{11}, Y_{12} respectively.

area we had available while respecting the waveguide spacing constraints for avoiding thermal crosstalk. More specifically, for the 4 × 4 GMZI, an input signal in port $j = 1$ would generate a pair of complementary outputs in output ports $k = 1$ and 4, with ER = 40 dB and FSR = 1.9 nm, where zero valued outputs are observed in the remaining ports. Additionally, for a single input at port $j = 2$, the previous behavior is alternated, with the resonant output pair being directed in ports $k = 2$ and 3 respectively. A similar behavior is also observed for the 3 × 3 GMZI, where depending on the input port choice, the resonant outputs (with FSR = 1.9 nm) are forming an on-resonance pair of different ER signals, complementary with the remaining output. For both configurations, the FSR is dictated by the approximately 600 um length difference of the waveguides.

However, the behavior of the fabricated devices is expected to deviate from the theoretical predictions. Three trivial types of errors are commonly encountered during fabrication and may affect the expected performance: errors in the arm lengths of the waveguides, phase errors in the MMI outputs and splitting ratio variations of the MMIs. The qualitative and quantitative effect of each of these errors on the GMZI response can be theoretically analyzed by taking into account certain variations of the waveguide length, phase and splitting ratio parameters around their targeted value.

Accounting, for example, the arm length variations in our model can be directly implemented by substituting the array waveguide matrix $\boldsymbol{\Phi}$ with a modified matrix $\boldsymbol{\Phi'}$, defined as

$$\boldsymbol{\Phi'} = \boldsymbol{\Phi} \times \boldsymbol{\delta L}. \tag{6}$$

where $\boldsymbol{\delta L}$ is a diagonal matrix with matrix elements $\delta L_{kk} = \exp[i\frac{2\pi n_{eff}}{\lambda}\delta x_k]$, with δx_k denoting the length variation.

Similarly, we can introduce the MMI phase variations, indicating the deviation of the acquired phase shifts at their output ports from the theoretically calculated values. We can introduce a modified MMI transfer matrix $\boldsymbol{S'}$ defined as:

$$\boldsymbol{S'} = \boldsymbol{S} \times \boldsymbol{\delta q}. \tag{7}$$

In (7), we have defined a complex matrix $\boldsymbol{\delta q}$, with matrix elements $\delta q_{jk} = \exp[i\frac{\lambda_0}{\lambda}\delta u_{jk}]$, where λ_0 is a reference wavelength equal to 1550 nm and δu_{jk} an arbitrary phase shift varying from $-\pi$ to $+\pi$.

Finally, variations in the splitting ratios of the MMIs can be incorporated in our analysis through a real diagonal matrix $\boldsymbol{\delta a}$, with elements a_{kk} close to unity, fulfilling the requirement that $\sum |a_{kk}|^2 = N$. In that manner, the MMI transfer matrix, including imbalanced splitting ratios will be equal to:

$$\boldsymbol{S'} = \boldsymbol{\delta a} \times \boldsymbol{S} \tag{8}$$

The indicative effects of all the specified independent variations to the characteristic outputs Y_{11}, Y_{12} for the 4 × 4 GMZI are summarized in Fig. 3. Fig. 3(a) and (b) depict the two 4 × 4 GMZI outputs when only arm 1 is varied, considering a δx_1 value in the range of -200 up to 200 nm. Fig. 3(c) and (d) illustrate the same outputs when only the MMI phase varies, considering $\boldsymbol{\delta q}$

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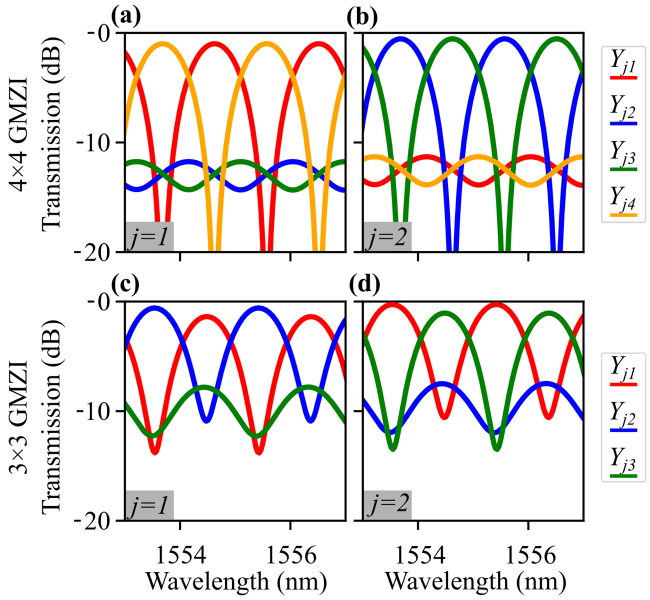


Fig. 4. Output spectra of 4 × 4 and 3 × 3 GMZI, under the simultaneous effect of multiple variations. For the 4 × 4 GMZI: $\delta x = (85, -85, 90, -40)$ nm $\delta u = (-\pi/5, \pi/7, 0, \pi/6)$ and $\delta a = (0.962, 1.012, 1.012, 0.962)$ and for the 3 × 3 GMZI: $\delta x = (90, -90, -110)$ nm $\delta u = (\pi/7, -\pi/6, \pi/7)$ and $\delta a = (0.987, 1.025, 0.987)$.

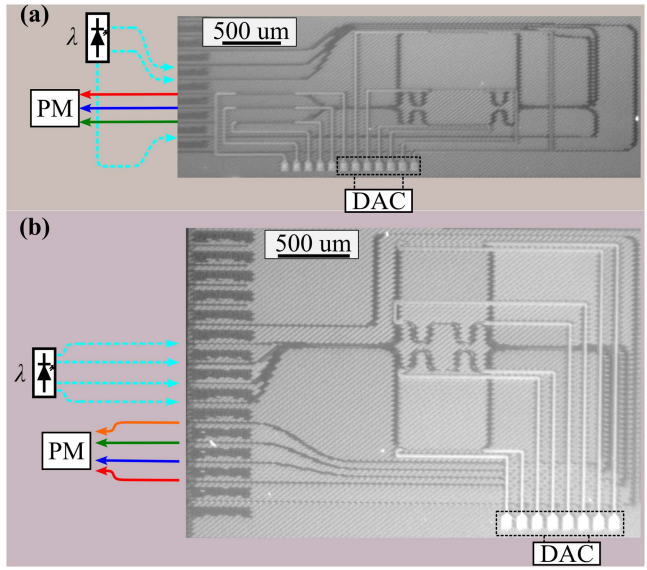


Fig. 5. The fabricated 3 × 3 (a) and 4 × 4 (b) GMZIs fabricated in Si_3N_4. A tunable laser is used for the generation of a CW signal, which is transmitted through a single input port of the GMZI and is recorded by a multichannel powermeter. Either the passive characterization of the devices can be performed, or a series of active measurements can be obtained, where the output of the GMZI is recorded at a single wavelength for different settings of the integrated thermo-optical phase shifters.

as a diagonal matrix with all elements being equal to 1, except from δu_{11} that is varied from $-\pi$ to π. The resonances emerging at Y_{11} (Fig. 3(a) and (c)), experience a wavelength shift and a change in the ER when either the arm length or the MMI phase vary. In the case of MMI phase error, the output resonances have also their peak power level decreasing with the error magnitude, as shown in Fig. 3(c). At the same time, the non-resonant GMZI outputs obtained at Y_{12} experience just a power-level shift in both cases, i.e., either in the case of an arm length variation or a phase error. Finally, the introduction of splitting ratio variations affects mostly the ER of both optical outputs, as shown in Fig. 3(e) and (f), with additional spectral shifting effects taking place at the inactive ports of the ideal 4 × 4 GMZI. Similar effects can be observed also in the 3 × 3 GMZI under the application of the same types of variations.

Fig. 4 depicts the outputs of both the 4 × 4 and 3 × 3 GMZI lossless configurations when random errors on all three parameters are simultaneously present. In particular, for the 4 × 4 GMZI we introduce the following variations: δx = (85,-85,90,-40) nm, δu = (-π/5,π/7,0,π/6) and δa = (0.962,1.012, 1.012,0.962) and for the 3 × 3 GMZI: δx = (90,-90,-110) nm and δu = (π/7,-π/6,π/7) and δa = (0.987,1.025,0.987). The presence of multiple random variations could give rise to shifts along the wavelength axis, and non-zero output power to the ideally vanishing outputs, justifying the complex response of each configuration.

III. Experimental Results

A. Device Fabrication and Experimental Setup

The Si_3N_4 3 × 3 and 4 × 4 GMZIs are presented in Fig. 5(a) and (b). The asymmetric devices have been fabricated in the low loss Si_3N_4 platform, with device footprint equal to 1.77 mm^2 and 2.7 mm^2 for the 3 × 3 and 4 × 4 configuration respectively. The cross section of the waveguides is set to 1 um. The waveguide losses in Si_3N_4 are expected to lie in the range of 0.1 dB/cm [20], characteristic of the platform's low loss credentials. The fiber- to-fiber losses, have been identified to 7.5 dB through single waveguide measurements, revealing a 3.7 dB loss per edge coupler. In order to tune the devices, thermo-optical phase shifters were employed in our design, with π-shift achieved for an applied electrical power in the range of 290 mW. A tunable laser is used as a continuous wave (CW) optical source, generating the single optical inputs X_j. The optical signal is forwarded to a polarization maintaining fiber array, to be injected to the GMZI through spot size converters. The optical outputs Y_{jk} of the GMZI are extracted through the same fiber array and are captured by a multichannel power-meter (PM). The integrated phase shifters are controlled by an electrical multi-channel digital to analog converter (DAC) through a multi-DC probe, allowing the simultaneous control of multiple thermo-optical phase shifters. In order to ensure the thermal and electrical stabilization of the phase shifters, which we identified that had a switching time lower than 10 ms, when alternating the applied electrical power, a stabilization time of 1.5 sec is introduced between sequential measurements.

We focus in two sets of measurements for both the 4 × 4 and the 3 × 3 GMZI devices. First, the passive characterization of the devices is performed by sweeping the wavelength of the CW source, and the normalized transmission of all possible outputs is recorded, in the absence of any induced phase shift. Subsequently, a series of measurements is performed for a single input wavelength and for different possible combinations of pairs of

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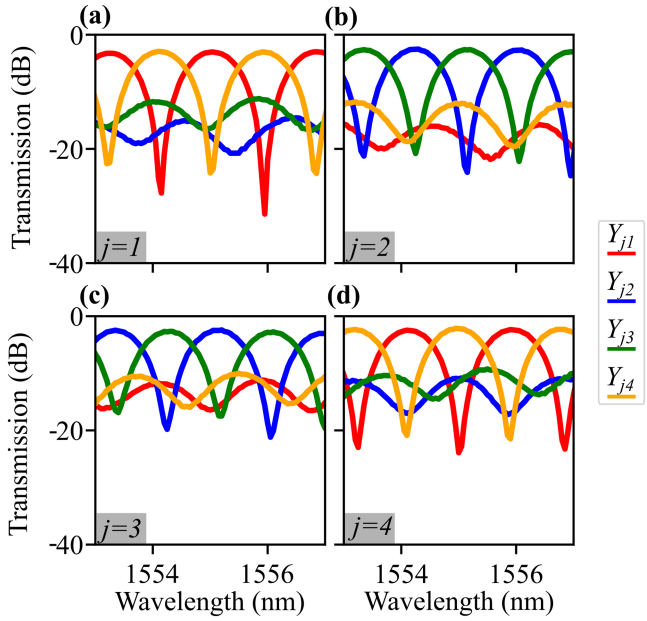


Fig. 6. Experimental output spectra of Si_3N_4 4×4 GMZI. The normalized transmission of all possible outputs under single input operation $X_1 - X_4$ are presented in (a)–(d) respectively.

phase shifters, which we attribute as the experimental GMZI configurations. Based on these measurements, we examine the performance of the devices as a switch and deduce the power splitting limits of each device.

In order to identify the optimal switch state from our measurements we follow an iterative procedure. For a specific output under consideration, the candidate switch states are identified as those that: i) have less than 7 dB of losses at the intended output ii) the crosstalk level at least 10 dB. For the case of multiple switch states fulfilling those criteria, the one with the minimum insertion loss is chosen. In the case that no switch state is extracted under those conditions, the crosstalk limit is iteratively reduced by steps of 0.5 dB, until at least a single switch state is identified.

Finally, the power splitting ratio limit of the devices can be extracted through our measurements. In particular, the power splitting ratio at each GMZI configuration for a specific output Y_{jk} is calculated by dividing the power obtained at the k-th output port by the total output power of the GMZI, which is calculated as the sum of the power values obtained at all GMZI output ports when the signal enters through input port j.

B. 4×4 GMZI

The normalized output power spectra of all outputs of the 4×4 GMZI are presented in Fig. 6, in the indicative range of 1553-1557 nm where an average insertion loss of approximately 2.4 ± 0.4 dB and an FSR of approximately 1.9 nm at all outputs is observed. Additionally, in agreement with our theoretical analysis, the complementary output pairs which contain the greatest percentage of the total output power are formed in our experimental measurements. However, clear differences are evident between the ideal theoretical model and the experimental results, such as the non-zero power values and the lack of complementarity to the ideally off-ports. Such deviations rise from the non-ideality of the device and can be justified by the inclusion of non-symmetric variations to the characteristics of the MMIs and the arrayed waveguides.

Additionally, we perform the active measurements of the 4×4 GMZI when the 1555 nm signal is injected from a single input port, with the indicative results concerning the switch operation and the power splitting limits of the device shown in Fig. 7, for the single inputs X_1 and X_2. The thermo-optical phase shifters are configured in pairs, with the applied electrical power ranging from 0 up to approximately 370 mW at each phase shifter, resulting to phase shifts greater than π and a total number of approximately 2300 different GMZI configurations. For this range of GMZI configurations, we observe that switch states with crosstalk in the range of 10 dB are identified, as shown in Fig. 7(a)–(d) and (i)–(l), revealing the adequate performance of the device as a switch. For example, the switch states at the outputs Y_{11}-Y_{14} were achieved for the $(\phi_1, \phi_2, \phi_3, \phi_4)$ GMZI phase shifter configurations of: (51,0,39,0), (300,0,192,0), (295,174,0,0) and (0,263,259,0) mW respectively, and the switch states at the outputs Y_{21}-Y_{24} were achieved for the $(\phi_1, \phi_2, \phi_3, \phi_4)$ GMZI phase shifter configurations of: (295,0,196,0), (0,247,252,0), (0,39,0,0) and (278,174,0,0) mW respectively. The behavior of the 4×4 GMZI could be further improved by varying three phase shifters simultaneously, which would however require additional independent channels in our electrical DAC.

Finally, the power splitting ratios, focusing at a specific single output at a time, for different GMZI configurations are presented in Fig. 7(e)–(h) and (m)–(p). It is evident that for all indicative outputs the splitting ratios have a maximum value that is always greater than 84%, with two of the four outputs always exhibiting unity splitting ratios. It should be mentioned that outputs Y_{11},Y_{14} and Y_{22},Y_{23} are expected to reach also unity splitting ratios if three phase shifters could be tuned simultaneously.

C. 3×3 GMZI

A similar set of measurements has been obtained for the 3×3 GMZI, where the relevant passive characterization of the device is presented in Fig. 8. A significantly lower average insertion loss of approximately 1 ± 0.2 dB is observed, with an FSR in the range of 1.9 nm also appearing in all outputs, again in full agreement with the theoretically expected FSR value for the specific arm length difference.

The active characterization of the device was performed with an input signal at 1554.7 nm. The phase shifters are tuned again in pairs, in a range of electrical power that ranged from 0 to approximately 270 mW. The relevant results are presented in Fig. 9, where the identified switch states and the experimentally obtained power splitting ratios for the single input X_1 (Fig. 9(a)–(f)) and X_2 (Fig. 9(g)–(l)) are presented. The switch states for the outputs Y_{11}-Y_{13} were obtained for the GMZI phase shifter configurations (34, 0, 91), (0, 248, 220) and (91, 156, 0) mW respectively and accordingly for the outputs Y_{21}-Y_{23} for the GMZI phase shifter configurations (227, 61, 0), (106, 178,

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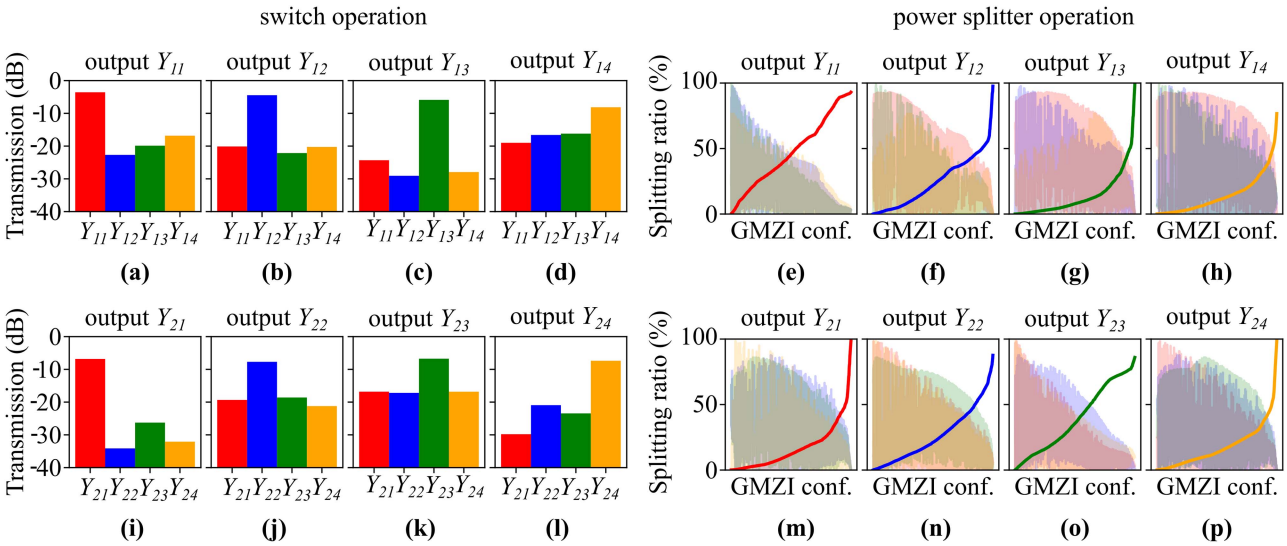


Fig. 7. Active measurements of 4 × 4 GMZI. (a)–(d): Switch operation of the 4 × 4 GMZI, with achieved crosstalk of at least 10 dB, for all the possible inputs when a single input X_1 is considered (e)–(f): Power splitting ratio for the same outputs of the GMZI, revealing the achievable upper power splitting ratio under the electrical power range applied to the phase shifters. The corresponding results for the single input X_2 are presented in (i)–(l) for the switch operation and (m)–(p) for the power splitting ratio respectively.

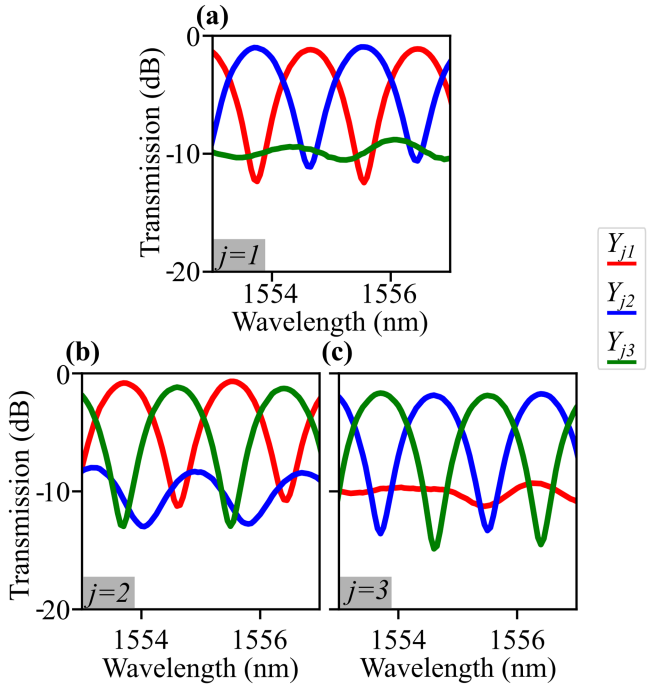


Fig. 8. Experimental output spectra of Si_3N_4 3 × 3 GMZI. The normalized transmission of all possible outputs under single input operation $X_1 - X_3$ are presented in (a)–(c) respectively.

0) and (24, 0, 68) mW respectively. The variation of pairs of phase shifters allows for the full control of the behavior of the 3 × 3 GMZI, as has been theoretically predicted, leading to experimental switching crosstalk levels in the range of 20 dB and greater than 12 dB when the signal enters through the X_1 and X_2, respectively, and always greater than 12 dB when the signal enters through X_3. At the same time, the power splitting ratio at all output ports extends along the entire range of [0,1]. Variation of the third phase shifter could improve further the obtained results, especially in switching, and under certain conditions reduce the overall power consumption of the device.V. GMZI based photonic architectures

The successful deployment of Si_3N_4-based GMZIs can allow for new architectural layouts that can eventually provide certain advantages over state-of-the-art designs that largely rely solely on examples in the areas of switching and unitary setups. Fig. 10(a) shows a non-blocking 4x4 switch that comprises an array of 2 × 2 MZI switches at its front- and back-end, i.e., before and after a 4 × 4 GMZI. The 4 × 4 non-blocking switch operation can be offered by 2 × 2 MZI-based designs through a 4 × 4 Benes architecture, requiring, however, a certain number of waveguide crossings in between its three switching stages [14] that may lead to differential path losses. The GMZI-based approach layout removes the need for waveguide crossings, being in principle capable of ensuring balanced path losses for any input-output port combination [19].

In the field of unitary optics, GMZIs structures form the fundamental building blocks in a recently proposed unitary optical circuit layout that is capable of supporting fidelity restoration [6]. This is a highly important property in linear optical circuits in order to enable the accurate experimental implementation of a targeted matrix, with state-of-the-art MZI-based unitary meshes inherently suffering from differential path losses and as such restricted to non-restorable fidelity setups [2], [3], [4]. This circuit design, shown in Fig. 10(b) for the case of a 4 × 4 setup, comprises a series of cascaded GMZIs ordered in increasing size and equipped with additional phase shifters at their outputs, forming in this way a universal GMZI (UGMZI). It has been shown that this scheme allows the realization of any unitary transformation [6] in the optical domain, and any real transformation when included in a Singular Value decomposition (SVD) scheme. With 2 × 2 MZIs being a well-established and low-loss building block in the Si_3N_4 waveguide platform, the 3 × 3 and 4 × 4 Si_3N_4-based GMZI structures demonstrated in this work can raise expectations for a 4 × 4 unitary optical circuit that can in principle allow for unity fidelity.

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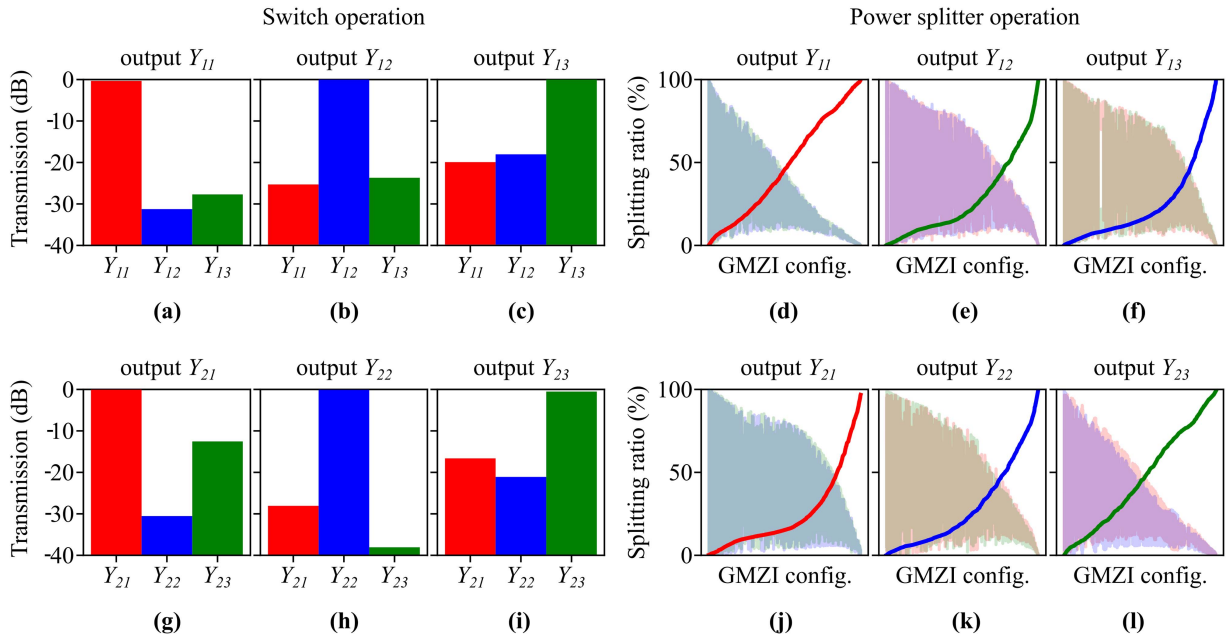


Fig. 9. Active measurements of 3 × 3 GMZI. (a)–(c): Switch operation of the 3 × 3 GMZI, with achieved crosstalk of at least 10 dB, for all the possible inputs when a single input X_1 is considered. (d)–(f): Power splitting ratios for the indicative outputs, revealing an almost 0–1 power splitting ratio range for all presented outputs. The corresponding results for input X_2 are presented in (g)–(h) for switching operation and in (j)–(l) for variable power splitting respectively.

TABLE I
GMZI IMPLEMENTATIONS IN DIFFERENT PLATFORMS

Technology	Application	Ports' Number	Insertion Loss (dB)	Crosstalk (dB)	Footprint (mm^2)
GaAs/AlGaAs [21]	Switching	10×10	12.2	-10	13×0.05
SOI [22]	Switching	3×3	N.A.	-11.1	1×0.08
Polymer (EPO) [23]	Switching	1×3	<5.5	<-19.7	26.7×0.07
InP [24]	Unitary	4×4	29.6	<-6.4	1.1×0.08
SiON [16]	Switching	4×4	N.A.	<-7	12
SiO_2 [25]	Switching	8×8	6	-34	N.A.
Si_3N_4	Unitary	2×2	0.13	N.A.	N.A.
Si_3N_4[this work]	Unitary/Switching	3×3 / 4×4	< 3	<-8	2.2×0.8/2×1.35

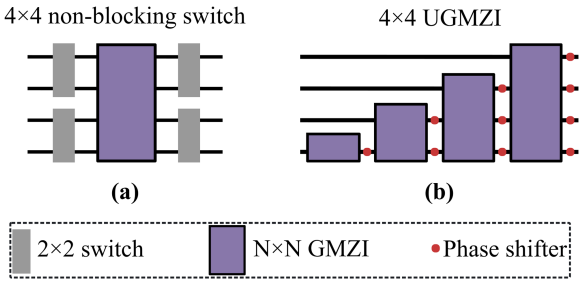


Fig. 10. Architectures encompassing GMZIs. (a) A 4 × 4 non-blocking switch, where the inclusion of the 4 × 4 GMZI allows the removal of all crossings. (b) A 4 × 4 UGMZI, consisting of a series of cascaded GMZIs of different size, equipped with phase shifters at their outputs, capable of constructing any arbitrary 4 × 4 unitary matrix.

IV. CONCLUSION

We have presented our analysis of Si_3N_4 3 × 3 and 4 × 4 GMZIs, where low insertion losses in the range of 1 and 2.4 dB per port have been observed respectively. Due to their tunability, we have presented their operation as switch modules, and variable power splitters, validating their versatile behavior and improved device footprint with respect to different photonic integration platforms, as presented in Table I. Further development and optimization of the designs of the GMZIs could pave the way for the experimental realization of alternative switching and linear optical circuits, with vast applications in emerging photonic technologies.

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